

1. Let  $V$  be a real inner product space. A set  $S = \{\mathbf{u}_1, \dots, \mathbf{u}_k\} \subseteq V$  is called *orthogonal* if it doesn't contain the zero vector, and if

$$\langle \mathbf{u}_i, \mathbf{u}_j \rangle = 0 \text{ for all } i \neq j.$$

(a) Let  $S = \{\mathbf{u}_1, \dots, \mathbf{u}_k\}$  be an orthogonal set and suppose that

$$\mathbf{v} = a_1 \mathbf{u}_1 + a_2 \mathbf{u}_2 + \dots + a_k \mathbf{u}_k.$$

In this case, show that  $a_i = \langle \mathbf{v}, \mathbf{u}_i \rangle / \langle \mathbf{u}_i, \mathbf{u}_i \rangle$  for all  $i$ . [Remark: If we also have  $\langle \mathbf{u}_i, \mathbf{u}_i \rangle = 1$  for all  $i$  then the set  $S$  is called *orthonormal*, in which case we have  $a_i = \langle \mathbf{v}, \mathbf{u}_i \rangle$ .]

(b) Use part (a) to prove that every orthogonal set is an independent set.

(a): Fix some  $1 \leq i \leq k$  and the inner product of each side with  $\mathbf{u}_i$  to get

$$\begin{aligned} \langle \mathbf{v}, \mathbf{u}_i \rangle &= \langle a_1 \mathbf{u}_1 + a_2 \mathbf{u}_2 + \dots + a_k \mathbf{u}_k, \mathbf{u}_i \rangle \\ &= a_1 \langle \mathbf{u}_1, \mathbf{u}_i \rangle + a_2 \langle \mathbf{u}_2, \mathbf{u}_i \rangle + \dots + a_k \langle \mathbf{u}_k, \mathbf{u}_i \rangle \\ &= a_i \langle \mathbf{u}_i, \mathbf{u}_i \rangle. \end{aligned}$$

Then divide both sides by  $\langle \mathbf{u}_i, \mathbf{u}_i \rangle$ , which is possible because  $\mathbf{u}_i \neq \mathbf{0}$ .

(b): Suppose that  $a_1 \mathbf{u}_1 + \dots + a_k \mathbf{u}_k = \mathbf{0}$ . Then part (a) tells us that

$$a_i = \frac{\langle \mathbf{0}, \mathbf{u}_i \rangle}{\langle \mathbf{u}_i, \mathbf{u}_i \rangle} = \frac{0}{\langle \mathbf{u}_i, \mathbf{u}_i \rangle} = 0 \quad \text{for all } i.$$

2. Let  $W$  be a vector space and let  $U, V \subseteq W$  be subspaces.

(a) Prove that  $U \cap V$  is a subspace, but  $U \cup V$  need not be a subspace.

(b) Define the sum of subspaces by

$$U + V = \{\mathbf{u} + \mathbf{v} : \mathbf{u} \in U, \mathbf{v} \in V\}.$$

Prove that  $U + V$  is a subspace of  $W$ .

(c) Suppose that  $U \cap V = \{\mathbf{0}\}$ . In this case, show that for all vectors  $\mathbf{w} \in U + V$  there exists a **unique** expression  $\mathbf{w} = \mathbf{u} + \mathbf{v}$  with  $\mathbf{u} \in U$  and  $\mathbf{v} \in V$ . [Hint: Suppose that  $\mathbf{u} + \mathbf{v} = \mathbf{u}' + \mathbf{v}'$  with  $\mathbf{u}, \mathbf{u}' \in U$  and  $\mathbf{v}, \mathbf{v}' \in V$ . Then ...]

(d) If  $W = U + V$  and  $U \cap V = \{\mathbf{0}\}$  then we say that  $W$  is the *direct sum* of  $U$  and  $V$ , and we write

$$W = U \oplus V.$$

In this case, show that  $\dim W = \dim U + \dim V$ . [Hint: Let  $\mathbf{u}_1, \dots, \mathbf{u}_m$  be a basis of  $U$  and let  $\mathbf{v}_1, \dots, \mathbf{v}_n$  be a basis of  $V$ . Prove that  $\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{v}_1, \dots, \mathbf{v}_n$  is a basis of  $W$ .]

(a): To show that  $U \cap V$  is a subspace, consider any  $\mathbf{x}, \mathbf{y} \in U \cap V$  and  $a, b \in \mathbb{R}$ . Since  $U \cap V \subseteq U$  we have  $\mathbf{x}, \mathbf{y} \in U$ , hence  $a\mathbf{x} + b\mathbf{y} \in U$  because  $U$  is a subspace. Similarly, since  $U \cap V \subseteq V$  and since  $V$  is a subspace we have  $a\mathbf{x} + b\mathbf{y} \in V$ . Hence  $a\mathbf{x} + b\mathbf{y} \in U \cap V$ , as desired. To show that  $U \cup V$  is **not** necessarily a subspace, pick any two linearly independent vectors  $\mathbf{x}, \mathbf{y} \in W$ , and consider the lines  $U = \text{Span}\{\mathbf{x}\} = \{a\mathbf{x} : a \in \mathbb{R}\}$  and  $V = \text{Span}\{\mathbf{y}\} = \{b\mathbf{y} : b \in \mathbb{R}\}$ , generated by  $\mathbf{x}$  and  $\mathbf{y}$ . Then the vector  $\mathbf{x} + \mathbf{y}$  is not in  $U$  because this would imply  $(1 - a)\mathbf{x} + \mathbf{y} = \mathbf{0}$  and it is not in  $V$  since this would imply  $\mathbf{x} + (1 - b)\mathbf{y} = \mathbf{0}$ , contradicting the independence of  $\mathbf{x}$  and  $\mathbf{y}$ . Hence  $\mathbf{x} + \mathbf{y}$  is not in  $U \cup V$ .

(b): Consider any vectors  $\mathbf{x}, \mathbf{y} \in U + V$  and scalars  $a, b \in \mathbb{R}$ . By definition we can write  $\mathbf{x} = \mathbf{u} + \mathbf{v}$  and  $\mathbf{y} = \mathbf{u}' + \mathbf{v}'$  for some  $\mathbf{u}, \mathbf{u}' \in U$  and  $\mathbf{v}, \mathbf{v}' \in V$ . Then then we have

$$a\mathbf{x} + b\mathbf{y} = a(\mathbf{u} + \mathbf{u}') + b(\mathbf{v} + \mathbf{v}') = (a\mathbf{u} + b\mathbf{u}') + (a\mathbf{v} + b\mathbf{v}'),$$

which is in  $U + V$  because  $a\mathbf{u} + b\mathbf{u}' \in U$  and  $a\mathbf{v} + b\mathbf{v}' \in V$ .

(c): Suppose that  $U \cap V = \{\mathbf{0}\}$  and suppose we have  $\mathbf{u} + \mathbf{v} = \mathbf{u}' + \mathbf{v}'$  for some  $\mathbf{u}, \mathbf{u}' \in U$  and  $\mathbf{v}, \mathbf{v}' \in V$ . This implies that  $\mathbf{u} - \mathbf{u}' = \mathbf{v}' - \mathbf{v}$ . But  $\mathbf{u} - \mathbf{u}' \in U$  and  $\mathbf{v}' - \mathbf{v} \in V$ , which implies that  $\mathbf{u} - \mathbf{u}'$  and  $\mathbf{v}' - \mathbf{v}$  are both in  $U \cap V$ . Since  $U \cap V = \{\mathbf{0}\}$  this implies that  $\mathbf{u} - \mathbf{u}' = \mathbf{0}$ , hence  $\mathbf{u} = \mathbf{u}'$ , and  $\mathbf{v}' - \mathbf{v} = \mathbf{0}$ , hence  $\mathbf{v} = \mathbf{v}'$ .

(d): Suppose that  $W = U + V$  and  $U \cap V = \{\mathbf{0}\}$ . Let  $\mathbf{u}_1, \dots, \mathbf{u}_m$  be a basis of  $U$  and let  $\mathbf{v}_1, \dots, \mathbf{v}_n$  be a basis of  $V$ . I claim that  $B = \{\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{v}_1, \dots, \mathbf{v}_n\}$  is a basis of  $W$  consisting of  $m + n$  distinct vectors, and hence

$$\dim W = m + n = \dim U + \dim V.$$

To see that the vectors are distinct, suppose that  $\mathbf{u}_i = \mathbf{v}_j$  for some  $i, j$ . But then  $\mathbf{u}_i, \mathbf{v}_j$  are both in  $U \cap V = \{\mathbf{0}\}$ , which implies  $\mathbf{u}_i = \mathbf{0}$  and  $\mathbf{v}_j = \mathbf{0}$ . But no basis can contain the zero vector. To show that  $B$  is a spanning set, note that for any  $\mathbf{w} \in W$  there exist  $\mathbf{u} \in U$  and  $\mathbf{v} \in V$  such that  $\mathbf{w} = \mathbf{u} + \mathbf{v}$  because we have  $W = U + V$  by assumption. Since  $\mathbf{u}_1, \dots, \mathbf{u}_m$  is a basis for  $U$  and  $\mathbf{v}_1, \dots, \mathbf{v}_n$  is a basis for  $V$ , there exist scalars  $a_i$  and  $b_j$  such that

$$\mathbf{w} = \mathbf{u} + \mathbf{v} = (a_1\mathbf{u}_1 + \dots + a_m\mathbf{u}_m) + (b_1\mathbf{v}_1 + \dots + b_n\mathbf{v}_n).$$

To show independence, suppose we have

$$a_1\mathbf{u}_1 + \dots + a_m\mathbf{u}_m + b_1\mathbf{v}_1 + \dots + b_n\mathbf{v}_n = \mathbf{0},$$

for some scalars  $a_i$  and  $b_j$ . This implies that

$$a_1\mathbf{u}_1 + \dots + a_m\mathbf{u}_m = -b_1\mathbf{v}_1 - \dots - b_n\mathbf{v}_n.$$

But the left hand side is a vector in  $U$  and the right hand side is a vector in  $V$ . Hence each side is in  $U \cap V = \{\mathbf{0}\}$ , which implies that each side equals  $\mathbf{0}$ . Then the independence of the  $\mathbf{u}_i$  implies that  $a_i = 0$  for all  $i$ , and the independence of the  $\mathbf{v}_j$  implies that  $b_j = 0$  for all  $j$ .

**3.** Let  $V$  be a real inner product space and let  $U \subseteq V$  be a subspace. We define the *orthogonal complement* of  $U$  as follows:

$$U^\perp := \{\mathbf{v} \in V : \langle \mathbf{v}, \mathbf{u} \rangle = 0 \text{ for all } \mathbf{u} \in U\}.$$

- (a) Prove that  $U^\perp$  is a subspace of  $V$ .
- (b) Prove that  $U \cap U^\perp = \{\mathbf{0}\}$ .
- (c) If  $V$  is finite dimensional, prove that  $V = U + U^\perp$ . Combining this with (b) gives

$$V = U \oplus U^\perp,$$

and hence  $\dim V = \dim U + \dim U^\perp$ . [Hint: Suppose that  $\dim U = k$  and  $\dim V = n$ , so that  $k \leq n$ . From the Gram-Schmidt procedure (see below), it is possible to find an orthonormal basis  $\mathbf{u}_1, \dots, \mathbf{u}_n$  for  $V$  such that  $\mathbf{u}_1, \dots, \mathbf{u}_k$  is a basis for  $U$ . Show that  $\mathbf{u}_{k+1}, \dots, \mathbf{u}_n$  are in  $U^\perp$  and use this to show that every  $\mathbf{v} \in V$  can be expressed as  $\mathbf{v} = \mathbf{u} + \mathbf{w}$  with  $\mathbf{u} \in U$  and  $\mathbf{w} \in U^\perp$ .]

(a): Consider any  $\mathbf{x}, \mathbf{y} \in U^\perp$  and  $a, b \in \mathbb{R}$ . Then for all  $\mathbf{u} \in U$  we have

$$\langle a\mathbf{x} + b\mathbf{y}, \mathbf{u} \rangle = a\langle \mathbf{x}, \mathbf{u} \rangle + b\langle \mathbf{y}, \mathbf{u} \rangle = a0 + b0 = 0,$$

which implies that  $a\mathbf{x} + b\mathbf{y} \in U^\perp$ .

(b): Suppose that  $\mathbf{x} \in U \cap U^\perp$ . Since  $\mathbf{x} \in U^\perp$  we have  $\langle \mathbf{x}, \mathbf{u} \rangle = 0$  for all  $\mathbf{u} \in U$ . But since  $\mathbf{x} \in U$  we can take  $\mathbf{u} = \mathbf{x}$  to obtain  $\langle \mathbf{x}, \mathbf{x} \rangle = 0$ . This implies  $\mathbf{x} = \mathbf{0}$  by the axioms of inner products.

(c): From the hint, we may assume that there exists an orthonormal basis  $\mathbf{u}_1, \dots, \mathbf{u}_n$  for  $V$  such that  $\mathbf{u}_1, \dots, \mathbf{u}_k$  is an orthonormal basis for  $U$ , for some  $0 \leq k \leq n$ .<sup>1</sup> For any  $1 \leq i \leq k$  and  $k+1 \leq j \leq n$  we have  $\langle \mathbf{u}_i, \mathbf{u}_j \rangle = 0$  by the assumed orthogonality of the basis. This implies that each  $\mathbf{u}_j$  with  $k+1 \leq j \leq n$  is orthogonal to each basis vector of  $U$ , hence is orthogonal to every vector in  $U$  because

$$\left\langle \sum_{i=1}^k a_i \mathbf{u}_i, \mathbf{u}_j \right\rangle = \sum_{i=1}^k a_i \langle \mathbf{u}_i, \mathbf{u}_j \rangle = 0.$$

The set  $\mathbf{u}_1, \dots, \mathbf{u}_n$  spans  $V$  by assumption, hence for any  $\mathbf{v} \in V$  we can write

$$\mathbf{v} = \sum_{i=1}^n a_i \mathbf{u}_i = \sum_{i=1}^k a_i \mathbf{u}_i + \sum_{j=k+1}^n a_j \mathbf{u}_j,$$

where  $\sum_{i=1}^k a_i \mathbf{u}_i \in U$  and  $\sum_{j=k+1}^n a_j \mathbf{u}_j \in U^\perp$ . Hence every  $\mathbf{v} \in V$  is in  $U + U^\perp$ .

**4.** Let  $V$  be a finite-dimensional real inner product space and let  $U \subseteq V$  be a subspace. From Problem 3(c) we know that  $V = U \oplus U^\perp$ . Hence from Problem 2(c) we know that for every vector  $\mathbf{x} \in V$  there exists a unique pair of vectors  $\mathbf{x}_U \in U$  and  $\mathbf{x}_{U^\perp} \in U^\perp$  such that

$$\mathbf{x} = \mathbf{x}_U + \mathbf{x}_{U^\perp}.$$

Define two functions  $P_U, P_{U^\perp} : V \rightarrow V$ , called the *orthogonal projections* onto  $U$  and  $U^\perp$ , by setting  $P_U(\mathbf{x}) := \mathbf{x}_U$  and  $P_{U^\perp}(\mathbf{x}) := \mathbf{x}_{U^\perp}$ . Let  $\mathbf{u}_1, \dots, \mathbf{u}_k$  be an orthogonal basis for  $U$ . In this case, show that

$$P_U(\mathbf{x}) = \frac{\langle \mathbf{x}, \mathbf{u}_1 \rangle}{\langle \mathbf{u}_1, \mathbf{u}_1 \rangle} \mathbf{u}_1 + \frac{\langle \mathbf{x}, \mathbf{u}_2 \rangle}{\langle \mathbf{u}_2, \mathbf{u}_2 \rangle} \mathbf{u}_2 + \dots + \frac{\langle \mathbf{x}, \mathbf{u}_k \rangle}{\langle \mathbf{u}_k, \mathbf{u}_k \rangle} \mathbf{u}_k$$

and

$$P_{U^\perp}(\mathbf{x}) = \mathbf{x} - P_U(\mathbf{x}) = \mathbf{x} - \sum_{i=1}^k \frac{\langle \mathbf{x}, \mathbf{u}_i \rangle}{\langle \mathbf{u}_i, \mathbf{u}_i \rangle} \mathbf{u}_i.$$

[Hint: Use Problem 1(a).]

Since  $\mathbf{u}_1, \dots, \mathbf{u}_k$  is a basis of  $U$  we can write  $\mathbf{x}_U = a_1 \mathbf{u}_1 + \dots + a_k \mathbf{u}_k$  for some scalars  $a_i$ , and Problem 1(a) tells us that  $a_i = \langle \mathbf{x}_U, \mathbf{u}_i \rangle / \langle \mathbf{u}_i, \mathbf{u}_i \rangle$ . But since  $\langle \mathbf{x}_{U^\perp}, \mathbf{u}_i \rangle = 0$  we have

$$\langle \mathbf{x}_U, \mathbf{u}_i \rangle = \langle \mathbf{x} - \mathbf{x}_{U^\perp}, \mathbf{u}_i \rangle = \langle \mathbf{x}, \mathbf{u}_i \rangle - \langle \mathbf{x}_{U^\perp}, \mathbf{u}_i \rangle = \langle \mathbf{x}, \mathbf{u}_i \rangle,$$

hence  $a_i = \langle \mathbf{x}, \mathbf{u}_i \rangle / \langle \mathbf{u}_i, \mathbf{u}_i \rangle$ .

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<sup>1</sup>Here we use the convention that  $V = V \oplus \{\mathbf{0}\}$ .

**Challenge Problem (Gram-Schmidt Procedure).** Let  $V$  be a finite-dimensional inner product space over  $\mathbb{R}$ . The *Gram-Schmidt procedure* is used to convert any basis  $\mathbf{v}_1, \dots, \mathbf{v}_n$  into an orthogonal basis  $\mathbf{u}_1, \dots, \mathbf{u}_n$ . We inductively define

$$\begin{aligned}\mathbf{u}_1 &:= \mathbf{v}_1, \\ \mathbf{u}_2 &:= \mathbf{v}_2 - \frac{\langle \mathbf{v}_2, \mathbf{u}_1 \rangle}{\langle \mathbf{u}_1, \mathbf{u}_1 \rangle} \mathbf{u}_1, \\ \mathbf{u}_3 &:= \mathbf{v}_3 - \frac{\langle \mathbf{v}_3, \mathbf{u}_1 \rangle}{\langle \mathbf{u}_1, \mathbf{u}_1 \rangle} \mathbf{u}_1 - \frac{\langle \mathbf{v}_3, \mathbf{u}_2 \rangle}{\langle \mathbf{u}_2, \mathbf{u}_2 \rangle} \mathbf{u}_2, \\ &\text{etc.}\end{aligned}$$

Afterwards, we may obtain an *orthonormal* basis by replacing each  $\mathbf{u}_i$  with  $\mathbf{u}_i / \|\mathbf{u}_i\|$ .

- (a) Use induction to prove that the following statement holds for all  $k \geq 1$ :

$$\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} = \text{Span}\{\mathbf{u}_1, \dots, \mathbf{u}_k\} \text{ and } \mathbf{u}_k \neq \mathbf{0}.$$

[Remark: We need  $\mathbf{u}_k \neq \mathbf{0}$  so we can divide by  $\langle \mathbf{u}_k, \mathbf{u}_k \rangle$  in the definition of  $\mathbf{u}_{k+1}$ .]

- (b) Use induction to prove that the following statement holds for all  $k \geq 1$ :

$$\langle \mathbf{u}_i, \mathbf{u}_k \rangle = 0 \quad \text{for all } i < k.$$

- (c) Define the subspace  $U_k := \text{Span}\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ . Use Problem 4 to show that

$$\mathbf{u}_{k+1} = P_{U_k^\perp}(\mathbf{v}_{k+1}) \quad \text{for all } k \geq 1.$$

The idea: We create a new vector  $\mathbf{u}_{k+1}$  that is perpendicular to the previous vectors  $\mathbf{u}_1, \dots, \mathbf{u}_k$  by projecting the original  $\mathbf{v}_{k+1}$  onto the orthogonal complement  $U_k^\perp$ . Parts (a) and (b) show that this idea works.

- (d) The Gram-Schmidt procedure can also be applied in infinite-dimensional vector spaces. Consider the function space  $L^2[0, 1]$  with inner product

$$\langle f(x), g(x) \rangle = \int_0^1 f(x)g(x) dx.$$

If we apply Gram-Schmidt to the infinite sequence  $1, x, x^2, \dots$ , then we obtain an infinite sequence of polynomials  $p_0(x), p_1(x), p_2(x), \dots$  called the *shifted Legendre polynomials*. Compute the first three Legendre polynomials and check that they are orthogonal with respect to the inner product.

- (e) Let  $U = \text{Span}\{1, x, x^2\} \subseteq L^2[0, 1]$ . Continuing from part (d), we know that

$$U = \text{Span}\{p_0(x), p_1(x), p_2(x)\}.$$

Use the formula from Problem 4 to compute the projection  $f(x) := P_U(x^3)$  onto the subspace  $U$ . Draw the graphs of  $x^3$  and  $f(x)$  on the same axes.

- (a): To save space we define the notation

$$\begin{aligned}U_k &:= \text{Span}\{\mathbf{u}_1, \dots, \mathbf{u}_k\}, \\ V_k &:= \text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\}.\end{aligned}$$

We will use induction to show that the following statement holds for all  $1 \leq k \leq n$ :

$$U_k = V_k \quad \text{and } \mathbf{u}_i \neq \mathbf{0} \text{ for } 1 \leq i \leq k.$$

The statement holds for  $k = 1$  because  $\mathbf{u}_1 = \mathbf{v}_1$  and  $\mathbf{v}_1 \neq \mathbf{0}$ . (No basis contains the zero vector.) Now suppose that the statement holds for some  $1 \leq k < n$ . In this case we need to show that  $U_{k+1} = V_{k+1}$  and  $\mathbf{u}_{k+1} \neq \mathbf{0}$ . Since  $\langle \mathbf{u}_i, \mathbf{u}_i \rangle \neq 0$  for all  $1 \leq i \leq k$  we may define

$$\mathbf{u}_{k+1} := \mathbf{v}_{k+1} - \sum_{i=1}^k \frac{\langle \mathbf{v}_{k+1}, \mathbf{u}_i \rangle}{\langle \mathbf{u}_i, \mathbf{u}_i \rangle} \mathbf{u}_i.$$

In order to show that  $V_{k+1} \subseteq U_{k+1}$  we will show that  $\mathbf{v}_1, \dots, \mathbf{v}_{k+1} \in U_{k+1}$ . We have  $\mathbf{v}_{k+1} \in U_{k+1}$  because

$$\mathbf{v}_{k+1} = \mathbf{u}_{k+1} + \sum_{i=1}^k \frac{\langle \mathbf{v}_{k+1}, \mathbf{u}_i \rangle}{\langle \mathbf{u}_i, \mathbf{u}_i \rangle} \mathbf{u}_i,$$

and we have  $\mathbf{v}_1, \dots, \mathbf{v}_k \in U_{k+1}$  because  $\mathbf{v}_1, \dots, \mathbf{v}_k \in V_k$  and  $V_k = U_k \subseteq U_{k+1}$ .

In order to show that  $U_{k+1} \subseteq V_{k+1}$  we will show that  $\mathbf{u}_1, \dots, \mathbf{u}_{k+1} \in V_{k+1}$ . We have  $\mathbf{u}_1, \dots, \mathbf{u}_k \in V_{k+1}$  because  $\mathbf{u}_1, \dots, \mathbf{u}_k \in U_k$  and  $U_k = V_k \subseteq V_{k+1}$ . And we have  $\mathbf{u}_{k+1} \in V_{k+1}$  because

$$\mathbf{w} := \sum_{i=1}^k \frac{\langle \mathbf{v}_{k+1}, \mathbf{u}_i \rangle}{\langle \mathbf{u}_i, \mathbf{u}_i \rangle} \mathbf{u}_i \in U_k = V_k \subseteq V_{k+1},$$

hence  $\mathbf{u}_{k+1} = \mathbf{v}_{k+1} - \mathbf{w} \in V_{k+1}$ . It remains to show that  $\mathbf{u}_{k+1} \neq \mathbf{0}$ . But if  $\mathbf{u}_{k+1} = \mathbf{0}$  then

$$\mathbf{v}_{k+1} = \sum_{i=1}^k \frac{\langle \mathbf{v}_{k+1}, \mathbf{u}_i \rangle}{\langle \mathbf{u}_i, \mathbf{u}_i \rangle} \mathbf{u}_i \in U_k = V_k,$$

which contradicts the fact that the set  $\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{v}_{k+1}$  is independent. (Any subset of an independent set is independent.)

(b): We have shown that  $U_n = V_n = V$ , so that  $\mathbf{u}_1, \dots, \mathbf{u}_n$  is a spanning set. To complete the proof we need to show that

$$\langle \mathbf{u}_i, \mathbf{u}_j \rangle = 0 \quad \text{for all } i \neq j.$$

We will do this by showing that the following statement holds for all  $1 \leq k \leq n$ :

$$\langle \mathbf{u}_i, \mathbf{u}_j \rangle = 0 \quad \text{for all } 1 \leq i < j \leq k.$$

For the case  $k = 1$  there is nothing to check. Suppose the statement holds for some  $1 \leq k < n$ , so that  $\langle \mathbf{u}_i, \mathbf{u}_j \rangle = 0$  for all  $1 \leq i < j \leq k$ . Then for any  $1 \leq i \leq k$  we also have

$$\begin{aligned} \langle \mathbf{u}_i, \mathbf{u}_{k+1} \rangle &= \left\langle \mathbf{u}_i, \mathbf{v}_{k+1} - \sum_{j=1}^k \frac{\langle \mathbf{v}_{k+1}, \mathbf{u}_j \rangle}{\langle \mathbf{u}_j, \mathbf{u}_j \rangle} \mathbf{u}_j \right\rangle \\ &= \langle \mathbf{u}_i, \mathbf{v}_{k+1} \rangle - \sum_{j=1}^k \frac{\langle \mathbf{v}_{k+1}, \mathbf{u}_j \rangle}{\langle \mathbf{u}_j, \mathbf{u}_j \rangle} \langle \mathbf{u}_i, \mathbf{u}_j \rangle \\ &= \langle \mathbf{u}_i, \mathbf{v}_{k+1} \rangle - \frac{\langle \mathbf{v}_{k+1}, \mathbf{u}_i \rangle}{\langle \mathbf{u}_i, \mathbf{u}_i \rangle} \langle \mathbf{u}_i, \mathbf{u}_i \rangle \\ &= 0. \end{aligned}$$

Thus  $\mathbf{u}_1, \dots, \mathbf{u}_n$  is an orthogonal basis for  $V$ .

(c): There is not much to show. Since  $\mathbf{u}_1, \dots, \mathbf{u}_k$  is an orthogonal basis for the subspace  $U_k \subseteq V$ , Problem 4 tells us that

$$P_{U_k^\perp}(\mathbf{v}_{k+1}) = \mathbf{v}_{k+1} - \sum_{i=1}^k \frac{\langle \mathbf{v}_{k+1}, \mathbf{u}_i \rangle}{\langle \mathbf{u}_i, \mathbf{u}_i \rangle} \mathbf{u}_i.$$

But the right hand side is exactly the definition of  $\mathbf{u}_{k+1}$ .

(d): Consider the subspace  $U = \text{Span}\{1, x, x^2\} \subseteq L^2[0, 1]$ . We can apply the Gram-Schmidt procedure to the basis  $1, x, x^2$  to obtain an orthogonal basis  $p_0(x), p_1(x), p_2(x)$  for  $U$  as follows. First we define

$$p_0(x) = 1.$$

Then we define

$$\begin{aligned} p_1(x) &= x - \frac{\langle x, p_0 \rangle}{\langle p_0, p_0 \rangle} p_0(x) \\ &= x - \frac{\int_0^1 x \, dx}{\int_0^1 1 \, dx} \cdot 1 \\ &= x - \frac{1}{2}. \end{aligned}$$

Next,

$$\begin{aligned} p_2(x) &= x^2 - \frac{\langle x^2, p_0 \rangle}{\langle p_0, p_0 \rangle} p_0(x) - \frac{\langle x^2, p_1 \rangle}{\langle p_1, p_1 \rangle} p_1(x) \\ &= x^2 - \frac{\int_0^1 x^2 \, dx}{\int_0^1 1 \, dx} \cdot 1 - \frac{\int_0^1 x^2(x - \frac{1}{2}) \, dx}{\int_0^1 (x - \frac{1}{2})^2 \, dx} \left(x - \frac{1}{2}\right) \\ &= x^2 - \frac{1}{3} - \frac{1/12}{1/12} \left(x - \frac{1}{2}\right) \\ &= x^2 - x + \frac{1}{6}. \end{aligned}$$

Thus the first three shifted Legendre polynomials are

$$p_0(x) = 1, \quad p_1(x) = x - \frac{1}{2}, \quad p_2(x) = x^2 - x + \frac{1}{6}.$$

We check that they are orthogonal:

$$\begin{aligned} \langle p_0, p_1 \rangle &= \int_0^1 \left(x - \frac{1}{2}\right) \, dx = 0, \\ \langle p_0, p_2 \rangle &= \int_0^1 \left(x^2 - x + \frac{1}{6}\right) \, dx = \frac{1}{3} - \frac{1}{2} + \frac{1}{6} = 0, \end{aligned}$$

and

$$\begin{aligned} \langle p_1, p_2 \rangle &= \int_0^1 \left(x - \frac{1}{2}\right) \left(x^2 - x + \frac{1}{6}\right) \, dx \\ &= \int_0^1 \left(x^3 - \frac{3}{2}x^2 + \frac{2}{3}x - \frac{1}{12}\right) \, dx \\ &= \frac{1}{4} - \frac{1}{2} + \frac{1}{3} - \frac{1}{12} \\ &= 0. \end{aligned}$$

(e): We can use the orthogonal basis  $p_0(x), p_1(x), p_2(x)$  for the space  $U$  of quadratic polynomials to compute the orthogonal projection of any function in  $L^2[0, 1]$  onto the subspace  $U$ . The projection  $P_U(f)$  is the quadratic polynomial that is closest to  $f$  in the  $L^2$  norm.

For the example  $f(x) = x^3$ , we have

$$P_U(x^3) = \frac{\langle x^3, p_0 \rangle}{\langle p_0, p_0 \rangle} p_0 + \frac{\langle x^3, p_1 \rangle}{\langle p_1, p_1 \rangle} p_1 + \frac{\langle x^3, p_2 \rangle}{\langle p_2, p_2 \rangle} p_2.$$

We compute

$$\frac{\langle x^3, p_0 \rangle}{\langle p_0, p_0 \rangle} = \frac{\int_0^1 x^3 dx}{\int_0^1 1 dx} = \frac{1}{4},$$

and

$$\frac{\langle x^3, p_1 \rangle}{\langle p_1, p_1 \rangle} = \frac{\int_0^1 x^3(x - \frac{1}{2}) dx}{\int_0^1 (x - \frac{1}{2})^2 dx} = \frac{3/40}{1/12} = \frac{9}{10}.$$

Finally,

$$\langle x^3, p_2 \rangle = \int_0^1 x^3 \left( x^2 - x + \frac{1}{6} \right) dx = \frac{1}{6} - \frac{1}{5} + \frac{1}{24} = \frac{1}{120},$$

and

$$\langle p_2, p_2 \rangle = \int_0^1 \left( x^2 - x + \frac{1}{6} \right)^2 dx = \frac{1}{180}.$$

Hence

$$\frac{\langle x^3, p_2 \rangle}{\langle p_2, p_2 \rangle} = \frac{1/120}{1/180} = \frac{3}{2}.$$

Therefore

$$\begin{aligned} f(x) := P_U(x^3) &= \frac{1}{4} + \frac{9}{10} \left( x - \frac{1}{2} \right) + \frac{3}{2} \left( x^2 - x + \frac{1}{6} \right) \\ &= \frac{3}{2}x^2 - \frac{3}{5}x + \frac{1}{20}. \end{aligned}$$

Thus

$$f(x) = \frac{3}{2}x^2 - \frac{3}{5}x + \frac{1}{20}.$$

Here are the graphs of  $x^3$  and  $f(x)$ :

